

CS471 – Lesson 22 Solutions Key

Dynamic Probabilistic Systems: Markov Models

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Theater Weather State Transition Model

State Space: $S = \{\text{Clear } (C), \text{Cloudy } (K), \text{Storm } (R)\}$.

a. **Transition Matrix T :**

Solution

$$T = \begin{pmatrix} 0.70 & 0.10 & 0.20 \\ 0.30 & 0.60 & 0.10 \\ 0.10 & 0.20 & 0.70 \end{pmatrix}$$

Row sums: $0.70 + 0.10 + 0.20 = 1.0$; $0.30 + 0.60 + 0.10 = 1.0$; $0.10 + 0.20 + 0.70 = 1.0$.

b. **Day 1 Probability Distribution (p_1):**

Solution

$$\mathbf{p}_1 = \mathbf{p}_0 T = (1.0 \quad 0.0 \quad 0.0) \begin{pmatrix} 0.70 & 0.10 & 0.20 \\ 0.30 & 0.60 & 0.10 \\ 0.10 & 0.20 & 0.70 \end{pmatrix} = \langle \mathbf{0.70}, \mathbf{0.10}, \mathbf{0.20} \rangle$$

c. **Day 2 Probability of Clear ($P(X_2 = C)$):**

Solution

Apply the mini-forward marginalization over intermediate states X_1 :

$$P(X_2 = C) = \sum_{i \in \{C, K, R\}} P(X_1 = i) \cdot P(X_2 = C \mid X_1 = i)$$

$$P(X_2 = C) = (0.70 \times 0.70) + (0.10 \times 0.30) + (0.20 \times 0.10)$$

$$P(X_2 = C) = 0.4900 + 0.0300 + 0.0200 = \mathbf{0.5400} \quad (54.00\%)$$

2. Assumptions of Markov Models

a. **First-Order Markov Assumption:**

Solution

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} \mid X_t)$$

Operational Meaning: The future state is conditionally independent of past history given the present state. An AI planner only needs to know current status (e.g., today's weather or current fuel level) to compute future distributions; historical states are irrelevant.

b. **Stationary Assumption (Time-Homogeneity):**

Solution

$$P(X_{t+1} = j \mid X_t = i) = P(X_1 = j \mid X_0 = i) \quad \text{for all } t$$

The transition dynamics are invariant over time.

Violation Scenario: Day/night diurnal heating cycles, seasonal shifts, or electronic warfare attacks that begin at a specific hour will alter transition probabilities over time, violating stationarity.

Part 2: Student Independent Practice Solutions

Tactical Satellite Communications (SATCOM) Degradation

Matrix:

$$T = \begin{pmatrix} 0.80 & 0.15 & 0.05 \\ 0.20 & 0.70 & 0.10 \\ 0.00 & 0.10 & 0.90 \end{pmatrix}, \quad \mathbf{p}_0 = \langle 1.0, 0.0, 0.0 \rangle$$

1. Multi-Step Probability Derivations:

Solution

a. **Distribution at $t = 1$:**

$$\mathbf{p}_1 = \mathbf{p}_0 T = \langle 1.0, 0.0, 0.0 \rangle T = \langle \mathbf{0.80}, \mathbf{0.15}, \mathbf{0.05} \rangle$$

b. **Probability Operational at $t = 2$ ($P(X_2 = O)$):**

$$P(X_2 = O) = \mathbf{p}_1 \cdot \text{Col}_1(T)$$

$$P(X_2 = O) = (0.80 \times 0.80) + (0.15 \times 0.20) + (0.05 \times 0.00)$$

$$P(X_2 = O) = 0.6400 + 0.0300 + 0.0000 = \mathbf{0.6700} \quad (67.00\%)$$

c. **Probability Severed at $t = 2$ ($P(X_2 = S)$):**

$$P(X_2 = S) = \mathbf{p}_1 \cdot \text{Col}_3(T)$$

$$P(X_2 = S) = (0.80 \times 0.05) + (0.15 \times 0.10) + (0.05 \times 0.90)$$

$$P(X_2 = S) = 0.0400 + 0.0150 + 0.0450 = \mathbf{0.1000} \quad (10.00\%)$$

Check: $P(X_2 = D) = (0.80)(0.15) + (0.15)(0.70) + (0.05)(0.10) = 0.1200 + 0.1050 + 0.0050 = 0.2300$. *Sum:* $0.67 + 0.23 + 0.10 = 1.0000$.

2. Trajectory Evaluation $P(X_0 = O, X_1 = D, X_2 = S)$:

Solution

$$P(X_0 = O, X_1 = D, X_2 = S) = P(X_0 = O) \cdot P(X_1 = D \mid X_0 = O) \cdot P(X_2 = S \mid X_1 = D)$$

$$P(X_0 = O, X_1 = D, X_2 = S) = 1.0 \times 0.15 \times 0.10 = \mathbf{0.0150} \quad (1.50\%)$$

3. Absorbing vs. Ergodic Behavior:

Solution

Yes, but only indirectly via Degraded (D).

Direct transition from $S \rightarrow O$ has probability 0.00 ($T_{31} = 0.0$). However, the satellite can transition from $S \rightarrow D$ with probability 0.10, and subsequently from $D \rightarrow O$ with probability 0.20. Therefore, the two-step probability of recovery is:

$$P(X_{t+2} = O \mid X_t = S) = (0.10 \times 0.20) = 0.0200 \quad (2.0\%)$$

State S is not an absorbing state because its self-loop is $0.90 < 1.0$.